Minimal Rank and Uniqueness of a Pole-Resolving Counterloop

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Abstract

We sharpen the resolution-clocked counterloop of RH-272. In the conjugate- symmetric class with zero odd moments, matching the even pole moments through order $2N$ requires at least $2N$ atoms. At equality the factor is uniquely $\Pi_N((\beta z)^2)$, hence its counterloop atoms are the nontrivial $(2N+2)$-th root directions on the pole circle. This is an algebraic minimality theorem for a graded counterterm, not a theorem about the noisy spectral cloud.

1 Setup

Let $\nu_1, \ldots, \nu_r$ be a finite multiset closed under conjugation and write $S_n = \sum_j \nu_j^n$. For a fixed $\beta > 0$, assume $S_{2m+1} = 0$ for $0 \le m \le N-1$ and $S_{2m} = -2\beta^{2m}$ for $1 \le m \le N$.

2 Theorem

Theorem 2.1 (minimal rank and equality)**.** *Under these assumptions $r \ge 2N$. If $r = 2N$, then*

$$\prod_{j=1}^{2N}(1 - z\nu_j) = \Pi_N((\beta z)^2)$$

and the multiset of counterloop atoms is $\{\beta e^{\pm ij\pi/(N+1)} : 1 \le j \le N\}$.

Proof. Put $F(z) = \prod_{j=1}^{r}(1 - z\nu_j)$. At the origin,

$$\log F(z) = -\sum_{n\ge1} \frac{S_n}{n} z^n.$$

The prescribed moments therefore make the Taylor series of $\log F$ agree through degree $2N$ with

$$\sum_{m=1}^{N} \frac{(\beta z)^{2m}}{m} = \log \Pi_N((\beta z)^2) + O(z^{2N+2}).$$

Exponentiation shows that F and $\Pi_N((\beta z)^2)$ have identical coefficients through degree $2N$. The latter has nonzero coefficient β^{2N} at degree $2N$, so $r \ge 2N$. If $r = 2N$, the two polynomials have the same degree and all their coefficients agree, hence they are equal. Factoring the geometric section gives the stated roots. □

3 Boundary

The theorem explains why the RH-272 rank clock is forced for a pole prefix. It does not provide a spectral selector for A_σ, and it does not upgrade finite noisy roots to this equality case. Gates A–E remain false/open.